

Associations between respiratory diseases and lung cancer risk: a secondary analysis of the large-scale prospective UK biobank cohort

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ABSTRACT

Introduction Respiratory diseases are significant risk factors for lung cancer; however, the association between acute respiratory infections and lung cancer incidence requires further exploration.

Methods We performed a secondary analysis of the prospective UK Biobank cohort, including participants aged 37–73 years recruited from 22 assessment centres across the UK between 2006 and 2010. Cox proportional hazards models estimated HRs for incident lung cancer according to respiratory disease status, which was defined based on linked hospital inpatient records. Mediation analysis explored potential biomarkers, and Mendelian randomisation assessed causal relationships.

Results During a mean follow-up of 10.44 years (4790 738 person-years), 2189 participants developed lung cancer. Among 107 007 individuals with respiratory diseases, 1322 cases occurred (incidence rate 27.6 per 10 000 person-years), compared with 867 cases among 351 876 participants without respiratory diseases (7.9 per 10 000 person-years). Overall, respiratory diseases were associated with increased lung cancer risk (HR 2.97, 95% CI 2.75 to 3.21). Acute respiratory infections, including acute nasopharyngitis (HR 3.41; 95% CI 1.83 to 6.34), influenza (HR 3.90; 95% CI 2.53 to 6.01), viral pneumonia (HR 9.86; 95% CI 6.55 to 14.85) and bacterial pneumonia (HR 6.28; 95% CI 4.40 to 8.96), showed strong associations with lung cancer incidence. In subtype analyses, squamous cell carcinoma exhibited the highest risk elevation (HR 3.65; 95% CI 3.06 to 4.36). Mediation analysis indicated that neutrophil counts partially mediated these associations (proportion mediated up to 8%).

Conclusion Acute respiratory infections were associated with higher lung cancer incidence, providing hypothesis-generating evidence that may inform future risk stratification research.

INTRODUCTION

Lung cancer remains a leading cause of cancer-related mortality worldwide, and respiratory diseases and infections contribute substantially to the global burden of morbidity and mortality.^{1–3} Chronic respiratory conditions (eg, COPD, tuberculosis and pulmonary fibrosis) have been linked

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Chronic respiratory diseases (eg, chronic obstructive pulmonary disease (COPD) and pneumonia) have been associated with subsequent lung cancer risk, whereas evidence for acute respiratory infections is less consistent.

WHAT THIS STUDY ADDS

⇒ In a large prospective UK Biobank cohort, a broad spectrum of respiratory diseases—including acute lower respiratory infections—was associated with a higher subsequent incidence of lung cancer.
⇒ In mediation analyses, baseline inflammation-related blood biomarkers accounted for only a modest proportion of these associations.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ These hypothesis-generating findings warrant replication in independent cohorts; they may eventually inform risk stratification research or evaluation in dedicated screening studies, but do not currently justify changes to established lung cancer screening criteria.

to elevated lung cancer risk,^{4–11} potentially through inflammation-related carcinogenic pathways,^{12–16} and acute respiratory infections have also been proposed to influence tumour development through immune dysregulation.¹⁷

However, although the link between chronic respiratory diseases and lung cancer is increasingly recognised, prior research on respiratory diseases and lung cancer has several important limitations. First, most studies have focused on chronic conditions, leaving uncertainty about whether and how acute respiratory episodes contribute to lung cancer risk.^{18–20} Evidence for pneumonia is particularly inconsistent: some studies report an increased risk of lung cancer after

pneumonia, with a pronounced elevation within the first year, whereas others have observed a lower subsequent incidence.^{21–25} These discrepant findings highlight ongoing debate regarding the pneumonia–lung cancer relationship and indicate a need for larger cohorts and more detailed investigation of potential mechanisms across lung cancer types.^{21–25} Second, research on upper respiratory tract diseases and lung cancer remains incomplete. While cohort studies have linked chronic rhinosinusitis to lung cancer,^{26,27} evidence for other upper-airway conditions (eg, pharyngeal or tonsillar diseases) is scarce. Overall, prior studies have often been limited by small sample sizes, short follow-up, residual confounding, uncertain causal interpretation and limited consideration of histological subtypes.^{21–28}

Acute respiratory infections, therefore, represent a common exposure of interest that may be associated with subsequent lung cancer incidence, yet the relationship remains uncertain.^{3,4,29} Recent reports during the COVID-19 pandemic suggest increased short-term lung cancer detection, likely reflecting intensified diagnostic activity.²⁹ To address these gaps, we used the prospective UK Biobank cohort to conduct an exploratory observational analysis using hospital-ascertained respiratory diagnoses, complemented by Mendelian randomisation (MR) sensitivity analyses. We aimed to describe the associations between a broad range of respiratory conditions—particularly acute infectious diseases—and subsequent lung cancer incidence across histological subtypes in this predominantly European population, and to explore whether inflammation-related biomarkers might partially mediate these relationships.

METHODS

Study population

This secondary analysis used data from the UK Biobank, a prospective cohort of 502 244 UK residents aged 37–73 years recruited between 2006 and 2010 across 22 assessment centres, with ethical approval obtained by UK Biobank/National Health Service (NHS).³⁰ Participants with prevalent lung cancer at baseline (n=23 745), <1 year of follow-up (n=17 336) or lung cancer diagnosed within 1 year after a respiratory disease diagnosis (n=2280) were excluded, leaving 458 883 participants for analysis (figure 1).

Assessment of respiratory disease and lung cancer

Incident respiratory disease was defined exclusively as the first recorded hospital inpatient diagnosis of respiratory system disorders (International Classification of Diseases (ICD)-10 J00–J99) using UK Biobank first-occurrence fields derived from linked NHS hospital records. Consequently, the analysis focuses on more severe disease episodes requiring hospitalisation, while diagnoses recorded only in primary care/outpatient settings were not treated as events. Codes were grouped into major categories (acute upper respiratory infections,

influenza/pneumonia/other acute lower respiratory infections, other upper respiratory tract diseases, chronic lower respiratory diseases, lung diseases due to external agents and other respiratory diseases; full code lists are in online supplemental table 1).

Incident lung cancer was identified through NHS cancer registry linkage using ICD-10 code C34. Histological subtypes were classified using ICD-O-3 morphology codes (details in online supplemental Methods).

Assessment of covariates

Baseline sociodemographic, lifestyle and anthropometric covariates were obtained from UK Biobank assessments, including age, sex, ethnicity, Townsend Deprivation Index (TDI), education, body mass index (BMI), waist-to-hip ratio (WHR), smoking status and pack-years (and years since quitting among former smokers), alcohol intake frequency, dietary factors (oily fish and processed meat intake) and family history of cancer. Variable definitions and categorisations are provided in online supplemental Methods. Missing indicators were used for categorical variables and median imputation for continuous variables.³¹

Assessment of potential mediators

Drawing from existing research on lung cancer, respiratory diseases and inflammatory markers, we identified 10 inflammation-related biomarkers as potential mediators.^{32–38} These blood-based biomarkers, subjected to rigorous quality assurance processes, have been externally validated. Importantly, all biomarkers were measured at the UK Biobank baseline assessment (Instance 0), collected at the time of participant recruitment between 2006 and 2010. In addition, the analysis required respiratory diseases to occur at least 1 year before lung cancer diagnosis. The selected markers include white blood cell count, neutrophil count, monocyte count, lymphocyte count, platelet count, C reactive protein and four systemic inflammatory indices: the lymphocyte-to-monocyte ratio, neutrophil-to-lymphocyte ratio, platelet-to-lymphocyte ratio and Systemic Immune-Inflammation Index (online supplemental Methods).

MR analysis

This study followed the Strengthening the Reporting of Observational Studies in Epidemiology-Mendelian randomisation (MR) guidelines.³⁹ Genome-wide association study (GWAS) summary statistics for respiratory diseases and lung cancer subtypes were used, with selected single nucleotide polymorphisms (SNPs) serving as instrumental variables (IVs) to infer causality. Since all data were obtained from publicly available GWAS datasets, no additional ethical approval was necessary.

In the two-sample MR analysis, genetic variants strongly associated with respiratory diseases but not directly linked to lung cancer were selected as IVs to reduce

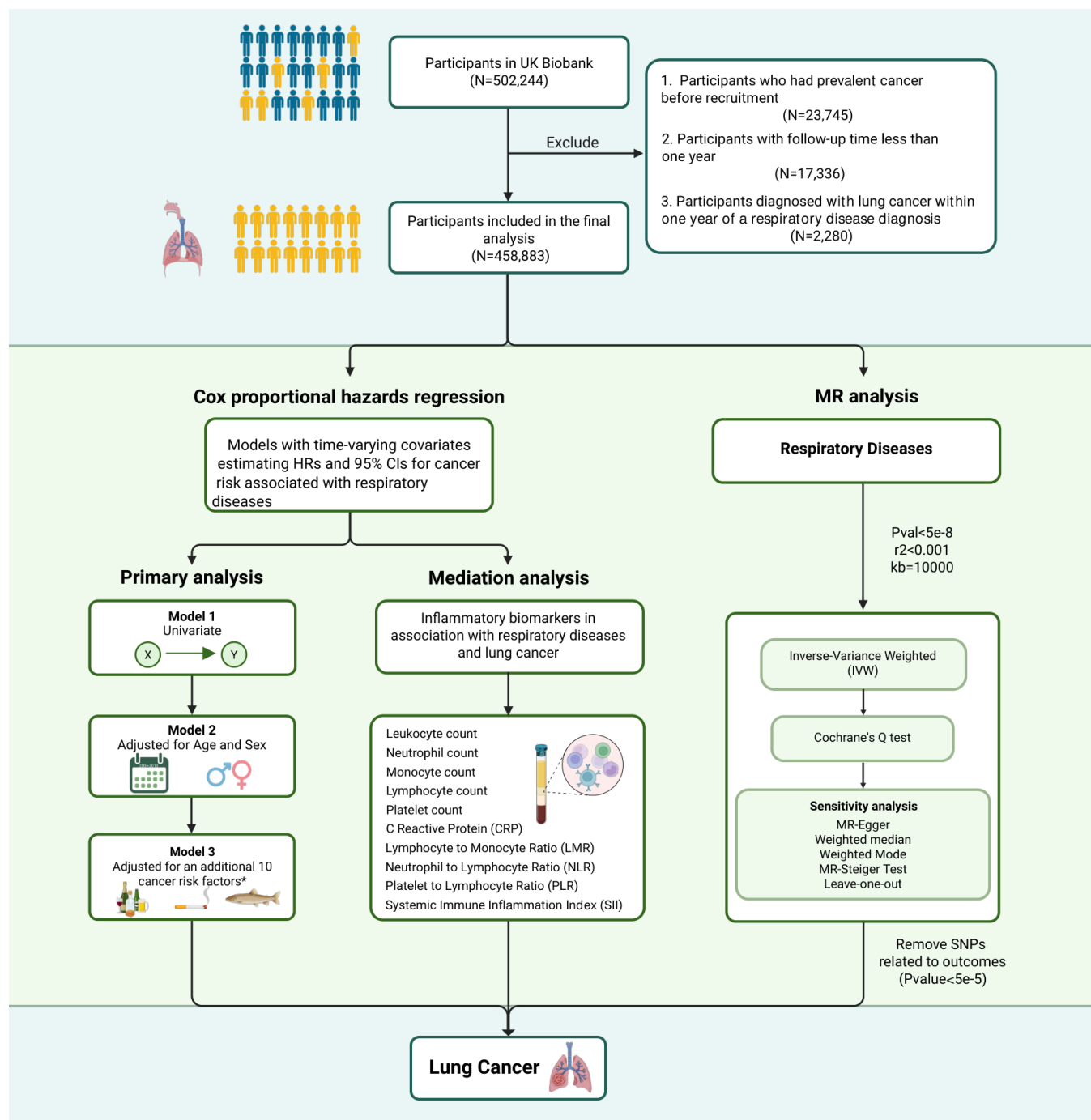


Figure 1 Study participants and study design. SNPs, single nucleotide polymorphisms.

confounding. The reliability of these IVs was verified using three independent databases.

Data on respiratory diseases were drawn from two UK Biobank sources: one analysed via the Scalable and Accurate Implementation of Generalized mixed model (SAIGE) method⁴⁰ and the other from UK Biobank (UKB) GWAS round 2⁴¹ (accessed on 1 August 2018, through Neale Lab). Lung cancer data were obtained from a meta-analysis involving organisations such as the International Agency for Research on Cancer, the Lunenfeld-Tanenbaum Research Institute and the University of Toronto.⁴² These summary statistics are publicly accessible

in the GWAS Catalogue (GCST004746, GCST004747, GCST004748, GCST004749 and GCST004750),⁴³ with further details in online supplemental tables 2 and 3.

The inverse variance-weighted (IVW) method was applied to estimate causal effects, with heterogeneity evaluated using Cochrane's Q test and I^2 statistics.⁴⁴ Fixed-effects IVW (IVW-FE) was used when heterogeneity was low ($I^2 < 25\%$) and mild ($I^2 < 50\%$),⁴⁵ while random-effects IVW (IVW-MRE) was employed in other cases.⁴⁶ Results were expressed as β values with SEs (figure 1).⁴⁷

Sensitivity analyses included MR-Egger regression to identify directional pleiotropy, with a significant

intercept indicating its presence. The weighted median method ensured robust causal estimates, even with up to 50% invalid SNPs,⁴⁸ while the weighted mode method assumed the largest cluster of estimates arose from valid instruments.⁴⁹ Horizontal pleiotropy was detected and corrected using Mendelian Randomisation Pleiotropy RESidual Sum and Outlier (MR-PRESSO) by identifying and excluding outliers.⁵⁰

A more detailed description of the MR procedures, including SNP selection criteria, linkage disequilibrium clumping thresholds, instrument strength evaluation and harmonisation steps, is provided in the online supplemental Methods. Summary measures of instrument strength (F-statistics, variance explained) and harmonisation details are presented in 10.5281/zenodo.18709849.⁵¹

Statistical analysis

Participants were followed from baseline to the earliest of lung cancer diagnosis, death, loss to follow-up or 1 June 2022, using the UK Biobank linkage snapshot generated in March 2024, which included linkage to national hospital inpatient, cancer and death registries up to this censoring date. Respiratory disease diagnoses had to precede any cancer diagnosis by at least 12 months. Lung cancer cases occurring within 1 year of a respiratory disease diagnosis were excluded. To ensure robust analysis, we focused on respiratory diseases with at least 10 lung cancer cases and 100 overall disease cases, resulting in the inclusion of 39 respiratory diseases as exposure factors.³¹

Time-varying Cox proportional hazard regression models were used to estimate HRs and 95% CIs for total and specific cancer risks, with follow-up time as the time scale. The “`timevary()`” function was applied to preprocess data, enabling modelling with time-varying covariates. Model 1 included univariate analysis, Model 2 adjusted for age and sex at baseline, and Model 3 incorporated additional covariates such as ethnicity, Townsend Deprivation Index, education level, BMI, WHR, smoking status, alcohol consumption, processed meat and oily fish intake and family cancer history. A sensitivity analysis that further adjusted for smoking pack-years and years since quitting smoking is provided in online supplemental table 16. For the primary analysis of overall lung cancer, Bonferroni correction was applied to the 37 primary respiratory diseases, as J99 and J100 represent distinct research questions and were therefore not included in the correction set. Sensitivity analyses excluded lung cancer cases diagnosed within 2 years or 6 months after respiratory disease onset, ensuring the robustness of findings. The Inverse Probability of Treatment Weighting Cox model was also used to analyse pooled data from all simulated trials (online supplemental table 6). We further verified the proportional hazards assumption and incorporated Fine–Gray competing-risk models to address the impact of death from other causes (online supplemental method).

To identify mediating inflammation markers, missing values were imputed using medians, and inflammation data were standardised as z-scores. Multiple linear regression assessed associations between respiratory diseases and inflammation markers, followed by Cox regression adjusted for model 3 covariates to evaluate links among respiratory diseases, inflammation markers and lung cancer. Mediating effects were quantified using the proportion mediated (PM).

All analyses were performed in R (V.4.4.1), with two-sided $p < 0.05$.

RESULTS

Baseline characteristics

This study included 458 883 participants, with an average follow-up period of 10.44 years (SD=3.47 years). Participants diagnosed with respiratory disease, in comparison to those without, were more likely to be male, smokers, have a family history of cancer and tended to be older (over 60 years). They are more likely to have a lower socioeconomic status, as measured by the Townsend Deprivation Index, exhibited lower WHRs, lower education levels, higher alcohol frequency, oily fish and processed meat intake (table 1).

Respiratory disease associated with lung cancer risk

Over 4 790 738 person-years of follow-up, 1322 lung cancer cases were identified among 107 007 patients with respiratory disease, while 867 cases occurred in 351 876 individuals without respiratory disease. After Bonferroni correction, in our model 3 analysis, we revealed that respiratory disease was associated with a significantly increased risk of lung cancer compared with participants without respiratory disease (HR 2.97, 95% CI 2.75 to 3.21, $p < 0.001$) (figure 2). We further examined associations with major histological subtypes of lung cancer. Compared with participants without respiratory disease, those with respiratory disease had higher risks of incident non-small cell lung cancer (NSCLC; HR 2.99, 95% CI 2.75 to 3.24) and small cell lung cancer (SCLC; HR 2.78, 95% CI 2.16 to 3.57). Within NSCLC, both lung adenocarcinoma (LUAD) and lung squamous cell carcinoma (LUSC)—histological subtypes of NSCLC—showed elevated risks (LUAD: HR 2.93, 95% CI 2.59 to 3.31; LUSC: HR 3.65, 95% CI 3.06 to 4.36; online supplemental figures 1–4). To further account for coexisting respiratory conditions, we grouped respiratory diseases into four broad categories based on ICD-10 codes and evaluated their individual associations with lung cancer risk. The results showed that influenza, pneumonia and other acute lower respiratory infections were associated with an increased risk of lung cancer (HR 1.23, 95% CI 1.08 to 1.41), as were chronic lower respiratory diseases such as COPD and asthma (HR 1.30, 95% CI 1.18 to 1.42). To account for multiple comparisons across four disease categories, we applied a Bonferroni correction (significance threshold $p < 0.125$). Checks of the proportional

Table 1 Baseline characteristics of participants according to the status of respiratory disease

Characteristic	Overall, n=458 883 (100%)*	Respiratory system diseases		P value†
		Yes, n=107 007 (23%)*	No, n=351 876 (77%)*	
Age (year), mean (SD)	58.00 (8.08)	60.00 (7.88)	57.00 (8.05)	<0.001
Age, n, %				<0.001
<60	263 016.00 (57.32%)	49 978.00 (46.72%)	213 038.00 (60.55%)	
≥60	195 806.00 (42.68%)	57 005.00 (53.28%)	138 801.00 (39.45%)	
Sex				<0.001
Male	210 423.00 (45.86%)	53 177.00 (49.71%)	157 246.00 (44.69%)	
Female	248 399.00 (54.14%)	53 806.00 (50.29%)	194 593.00 (55.31%)	
Ethnicity, n, %				<0.001
White	431 800.00 (94.11%)	100 304.00 (93.76%)	331 496.00 (94.22%)	
Mixed	2628.00 (0.57%)	615.00 (0.57%)	2013.00 (0.57%)	
Asian or Asian British	8983.00 (1.96%)	2322.00 (2.17%)	6661.00 (1.89%)	
Black or Black British	6912.00 (1.51%)	1608.00 (1.50%)	5304.00 (1.51%)	
Other ethnic group	5402.00 (1.18%)	1215.00 (1.14%)	4187.00 (1.19%)	
Unknown	3097.00 (0.67%)	919.00 (0.86%)	2178.00 (0.62%)	
Townsend Deprivation Index, mean (SD)	−2.15 (3.08)	−1.74 (3.28)	−2.26 (3.00)	<0.001
Townsend Deprivation Index, n, %				<0.001
<−2.14	230 300.00 (50.19%)	48 244.00 (45.10%)	182 056.00 (51.74%)	
≥−2.14	228 522.00 (49.81%)	58 739.00 (54.90%)	169 783.00 (48.26%)	
BMI, mean (SD)	26.71 (5.22)	27.54 (5.97)	26.48 (4.95)	<0.001
BMI, n, %				<0.001
<25	153 337.00 (33.42%)	29 653.00 (27.72%)	123 684.00 (35.15%)	
25–29.9	193 768.00 (42.23%)	43 849.00 (40.99%)	149 919.00 (42.61%)	
≥30	111 717.00 (24.35%)	33 481.00 (31.30%)	78 236.00 (22.24%)	
Waist-to-hip ratio, mean (SD)	0.87 (0.09)	0.90 (0.09)	0.87 (0.09)	<0.001
Waist-to-hip ratio, n, %				<0.001
<0.87	222 577.00 (48.51%)	42 373.00 (39.61%)	180 204.00 (51.22%)	
≥0.87	236 245.00 (51.49%)	64 610.00 (60.39%)	171 635.00 (48.78%)	
Education, n, %				<0.001
No college degree	304 977.00 (66.47%)	77 977.00 (72.89%)	227 000.00 (64.52%)	
College or university degree	149 015.00 (32.48%)	27 486.00 (25.69%)	121 529.00 (34.54%)	
Unknown	4830.00 (1.05%)	1520.00 (1.42%)	3310.00 (0.94%)	
Smoking status, n, %				<0.001
Never	250 944.00 (54.69%)	49 408.00 (46.18%)	201 536.00 (57.28%)	
Previous	157 801.00 (34.39%)	41 447.00 (38.74%)	116 354.00 (33.07%)	
Current	47 539.00 (10.36%)	15 314.00 (14.31%)	32 225.00 (9.16%)	
Prefer not to answer	2538.00 (0.55%)	814.00 (0.76%)	1724.00 (0.49%)	
Family cancer, n, %				<0.001
No	311 476.00 (67.89%)	72 084.00 (67.38%)	239 392.00 (68.04%)	
Yes	139 491.00 (30.40%)	32 677.00 (30.54%)	106 814.00 (30.36%)	
Unknown	7855.00 (1.71%)	2222.00 (2.08%)	5633.00 (1.60%)	
Alcohol frequency, n, %				<0.001
Never	36 412.00 (7.94%)	11 325.00 (10.59%)	25 087.00 (7.13%)	

Continued

Table 1 Continued

Characteristic	Overall, n=458 883 (100%)*	Respiratory system diseases		P value†
		Yes, n=107 007 (23%)*	No, n=351 876 (77%)*	
Special occasion	52 240.00 (11.39%)	14 695.00 (13.74%)	37 545.00 (10.67%)	
1–3 times a month	51 039.00 (11.12%)	11 816.00 (11.04%)	39 223.00 (11.15%)	
1–2 times a week	118 074.00 (25.73%)	25 980.00 (24.28%)	92 094.00 (26.18%)	
3–4 times a week	106 343.00 (23.18%)	21 604.00 (20.19%)	84 739.00 (24.08%)	
Almost every day	93 446.00 (20.37%)	21 123.00 (19.74%)	72 323.00 (20.56%)	
Unknown	1268.00 (0.28%)	440.00 (0.41%)	828.00 (0.24%)	
Oily fish intake, n, %				<0.001
Never	50 099.00 (10.92%)	13 212.00 (12.35%)	36 887.00 (10.48%)	
<1 time a week	151 192.00 (32.95%)	33 791.00 (31.59%)	117 401.00 (33.37%)	
1 time a week	172 125.00 (37.51%)	38 959.00 (36.42%)	133 166.00 (37.85%)	
2–4 times a week	77 531.00 (16.90%)	18 764.00 (17.54%)	58 767.00 (16.70%)	
5–6 times a week	3226.00 (0.70%)	833.00 (0.78%)	2393.00 (0.68%)	
Almost every day	1082.00 (0.24%)	273.00 (0.26%)	809.00 (0.23%)	
Unknown	3567.00 (0.78%)	1151.00 (1.08%)	2416.00 (0.69%)	
Processed meat intake, n, %				<0.001
Never	42 478.00 (9.26%)	8969.00 (8.38%)	33 509.00 (9.52%)	
<1 time a week	139 229.00 (30.34%)	30 568.00 (28.57%)	108 661.00 (30.88%)	
1 time a week	133 374.00 (29.07%)	30 894.00 (28.88%)	102 480.00 (29.13%)	
2–4 times a week	123 715.00 (26.96%)	30 969.00 (28.95%)	92 746.00 (26.36%)	
5–6 times a week	14 376.00 (3.13%)	3887.00 (3.63%)	10 489.00 (2.98%)	
Almost every day	3740.00 (0.82%)	1068.00 (1.00%)	2672.00 (0.76%)	
Unknown	1910.00 (0.42%)	628.00 (0.59%)	1282.00 (0.36%)	

*mean (SD) for continuous; n (%) for categorical.
†Wilcoxon rank sum test; Pearson's Chi-squared test.
BMI, body mass index.

hazards assumption are shown in online supplemental table 17 and online supplemental figure 7, whereas the results of Fine–Gray competing-risk models and cumulative incidence function curves by major respiratory disease groups are provided in online supplemental table 18 and online supplemental figure 8.

Acute respiratory infections associated with lung cancer

Acute respiratory infections are significantly associated with an increased risk of lung cancer (p value after Bonferroni correction) compared with those without respiratory diseases, which included acute nasopharyngitis (HR 3.41; 95% CI 1.83 to 6.34), acute pharyngitis (HR 3.45; 95% CI 2.19 to 5.42), influenza due to identified influenza virus (HR 3.90; 95% CI 2.53 to 6.01), viral pneumonia (HR 9.86; 95% CI 6.55 to 14.85), pneumonia due to *Streptococcus pneumoniae* (HR 4.83; 95% CI 2.90 to 8.05), bacterial pneumonia (HR 6.28; 95% CI 4.40 to 8.96), pyothorax (HR 5.34; 95% CI 3.21 to 8.88) and pleural effusion (HR 13.77; 95% CI 8.81 to 21.52) (figure 2).

Sensitivity analysis

In the sensitivity analysis, when only participants who had not been diagnosed with respiratory diseases at baseline were included, a consistent association between respiratory diseases and the risk of lung cancer was observed. In the sensitivity analysis focusing on lung cancer subtypes, significant links were observed between respiratory diseases and NSCLC, adenocarcinoma, as well as squamous cell carcinoma. This outcome underscores the robustness of the findings (online supplemental tables 4 and 5). Additionally, a weighted Cox analysis was conducted on a pseudo dataset created through inverse probability weighting in simulation experiments. The analysis revealed a stable association between respiratory diseases and lung cancer risk (online supplemental table 6).

To further address the possibility of reverse causation, we also examined longer latency periods for five respiratory conditions with particularly elevated estimates—viral pneumonia, not elsewhere classified; bacterial pneumonia, not elsewhere classified; pneumonia, organism

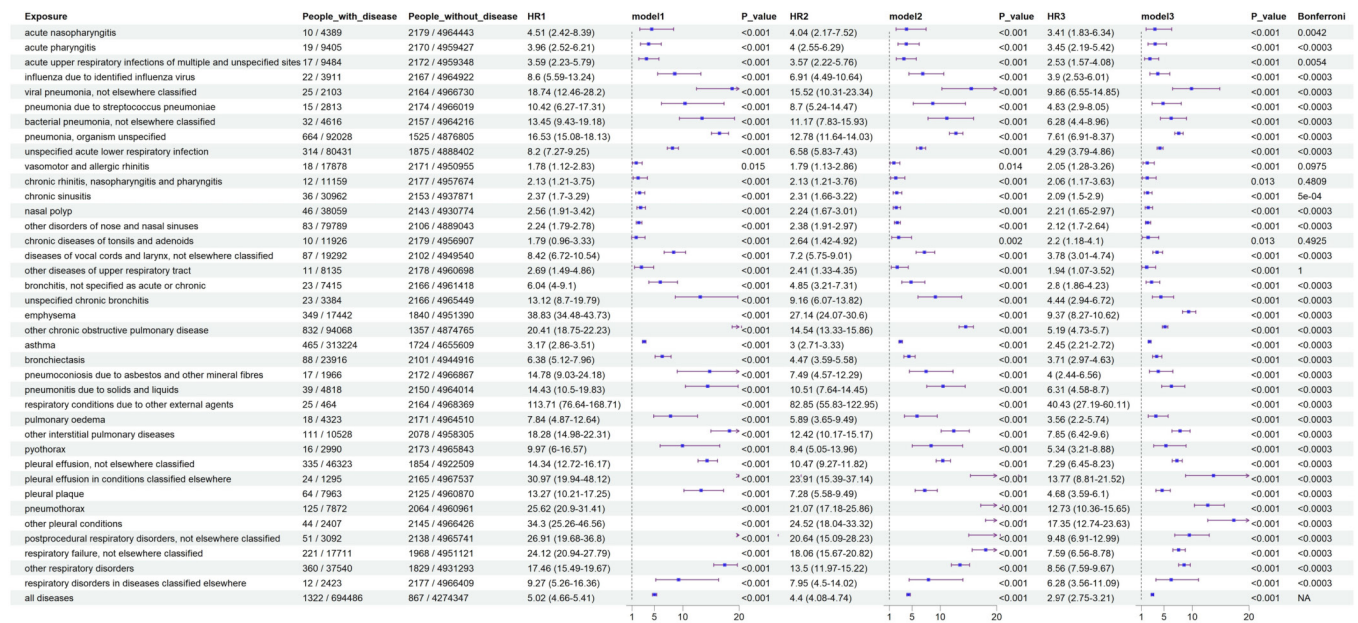


Figure 2 Respiratory diseases associated with lung cancer. ‘People with diseases’ and ‘People without diseases’ represent lung cancer cases/person-years in the exposed and unexposed groups, respectively. Model 1 is a univariate Cox analysis. Model 2 is adjusted for age and sex. Model 3 is adjusted for age, BMI, Townsend Deprivation Index, waist-to-hip ratio, sex, ethnicity, family history of cancer, education level, smoking status, alcohol consumption frequency, oily fish intake and processed meat intake.

unspecified; pleural effusion, not elsewhere classified and pleural effusion in conditions classified elsewhere. HRs were recalculated after sequentially excluding participants who developed lung cancer within 1, 2, 3, 4 and 5 years of the respiratory diagnosis (online supplemental figure 5 and online supplemental table 15).

Subgroup analysis

To analyse the association between respiratory diseases and lung cancer in different subpopulations, we conducted subgroup analyses on various covariates, including age, sex, ethnicity, TDI, BMI, WHR, education level, smoking status, family history of cancer, alcohol consumption, intake of oily fish and intake of processed meats.

In the sex-stratified analysis, the risk of lung cancer in males (HR 3.17, 95% CI 2.84 to 3.53) was found to be potentially higher than in females (HR 2.70, 95% CI 2.41 to 3.02). In the alcohol stratification analysis, we observed that as the frequency of alcohol consumption increased, the risk of developing lung cancer correspondingly increased: participants who drank almost every day (HR 3.15, 95% CI 2.67 to 3.71), followed by those who drank 1–3 times a month to 3–4 times a week (HR 2.91, 95% CI 2.64 to 3.21), and participants who never drank alcohol (HR 2.65, 95% CI 2.14 to 3.28) (figure 3).

Stratified analyses by smoking status (never, former, current) were additionally conducted across major lung cancer histological subtypes, including NSCLC, SCLC, LUAD and LUSC, with results shown in online supplemental figure 6. Within the NSCLC subtype, former

smokers demonstrated the highest relative risk associated with respiratory diseases (HR 3.37, 95% CI 2.98 to 3.81), compared with current smokers (HR 2.61, 95% CI 2.30 to 2.96) and never smokers (HR 2.53, 95% CI 1.96 to 3.26).

Mediation analysis

We analysed the PM of 10 mediating biomarkers between 38 respiratory diseases and lung cancer. The impact of individual respiratory diseases on the increased risk of lung cancer was partially attributed to inflammation-related biomarkers, with PM ranging from –0.01 to 0.08. Among all the inflammation biomarkers analysed, the mediation effects of neutrophils were more pronounced (figure 4). Detailed associations between inflammation-related biomarkers, respiratory diseases and lung cancer can be seen in the 10.5281/zenodo.18709849.

DISCUSSION

In this exploratory analysis of a large prospective cohort from the UK Biobank, we observed that respiratory diseases were associated with an increased incidence of lung cancer. Previous epidemiological studies have found that many chronic respiratory diseases increase the risk of lung cancer incidence, and our study confirms this finding.^{3–11 52} Compared with previous studies, our work adds a more comprehensive description of these associations across a broad spectrum of respiratory conditions, including acute infections, within a single population-based dataset. As expected, HRs were smaller after grouping diagnoses by the first qualifying respiratory

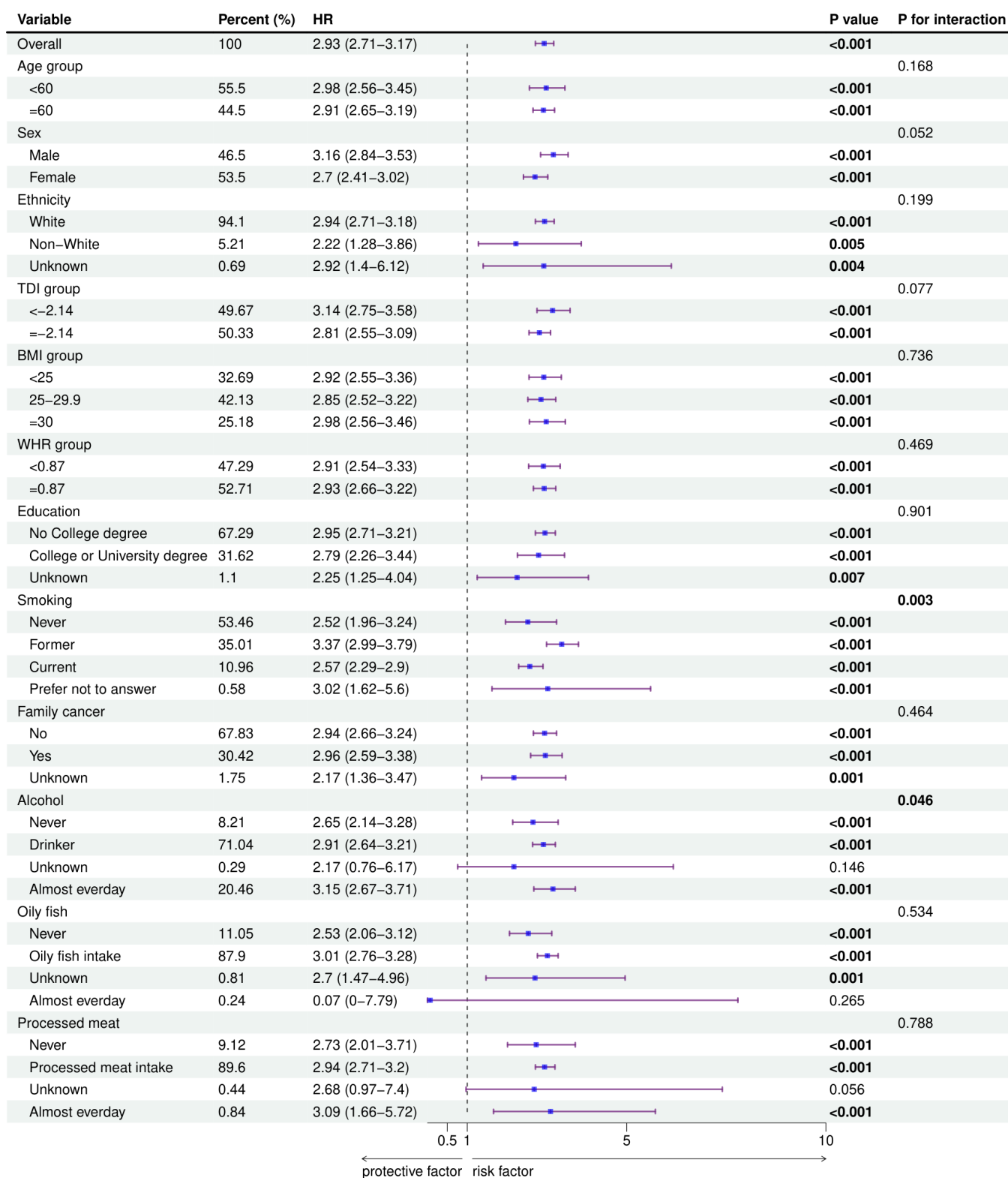


Figure 3 Subgroup analysis of respiratory diseases associated with lung cancer. The Cox proportional hazards model was adjusted for age, body mass index (BMI), Townsend Deprivation Index, waist-to-hip ratio (WHR), sex, ethnicity, family history of cancer, education level, smoking status, alcohol consumption frequency, oily fish intake and processed meat intake.

condition, which better accounts for coexisting diseases and increases sample size, yielding more stable estimates. These findings may help to deepen our understanding of the epidemiology of respiratory diseases and lung cancer,

but they should be interpreted as hypothesis-generating rather than definitive evidence of causality. Importantly, they do not justify a change in clinical practice or screening protocols; instead, they provide a rationale for

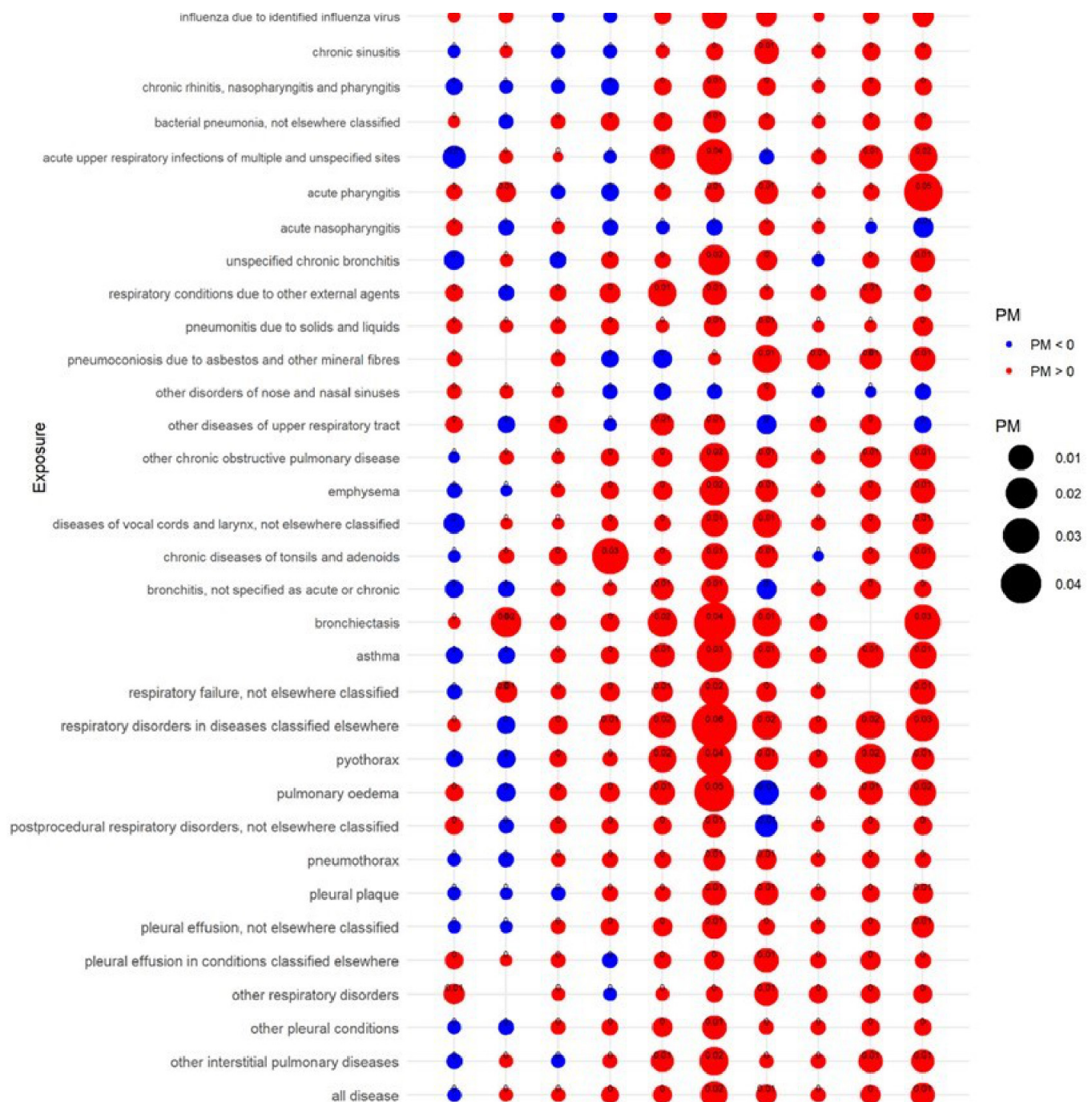


Figure 4 Proportion mediated (PM) of mediating biomarkers between respiratory diseases and lung cancer. The Cox proportional hazards model was adjusted for age, body mass index (BMI), Townsend Deprivation Index, waist-to-hip ratio, sex, ethnicity, family history of cancer, education level, smoking status, alcohol consumption frequency, intake of oily fish and intake of processed meat.

evaluating these associations in future research focused on refined risk evaluation. In subgroup analysis, respiratory diseases were associated with multiple lung cancer subtypes (NSCLC, SCLC, LUAD and LUSC), with the strongest elevation observed for lung squamous cell carcinoma.

Regarding the underlying mechanisms of this phenomenon, chronic respiratory diseases may increase lung cancer risk through a sustained proinflammatory

microenvironment.^{12–14} This environment generates oxidative stress that damages DNA or induces apoptosis, leading to the accumulation of mutations and an elevated cancer risk.^{15–17} Socioeconomic deprivation may further heighten lung cancer risk through higher smoking rates and greater exposure to urban air pollution. Recent evidence shows that PM_{2.5} can promote carcinogenesis by enhancing inflammatory expansion of

pre-existing mutant clones, providing additional environmental context to the findings.⁵⁴

For acute respiratory infections, the epidemiological literature remains limited and sometimes conflicting. To our knowledge, associations of acute nasopharyngitis and acute pharyngitis with subsequent lung cancer have not been well characterised in prior population studies and therefore require replication. For pneumonia, prior reports have described higher short-term detection and/or higher incidence of lung cancer after pneumonia, while other studies have suggested null or inverse associations.^{21–24} In this study, we observed that several acute respiratory infections—including acute nasopharyngitis, acute pharyngitis, influenza and pneumonia phenotypes—were associated with higher lung cancer incidence, but this observed association is hypothesis-generating and needs replication with longer follow-up and clearer temporal separation.⁵⁵

Biologically, acute infections may transiently disrupt the airway epithelial barrier and impair tissue repair, potentially increasing vulnerability to inhaled carcinogens.^{55–57} Findings from COVID-19 provide relevant context: SARS-CoV-2 infection can cause substantial epithelial/endothelial injury and prolonged inflammatory perturbations, which could, in theory, contribute to tissue vulnerability after recovery. At the same time, the COVID-19 era also plausibly increased the detection of subclinical lung cancers due to higher rates of chest imaging during and after acute illness, raising the possibility of detection bias for acute infections and pleural conditions.^{17 58–61} Mechanistically, acute infection-triggered inflammation has been proposed to involve cytokine signalling (eg, tumor necrosis factor (TNF)- α) and downstream pathways such as Janus kinase (JAK) 1/2–signal transducer and activator of transcription (STAT)1, among others, which could reshape the pulmonary microenvironment and influence epithelial survival and repair. However, whether these pathways mediate the causal effect of acute infections on lung cancer initiation or progression remains uncertain and requires further experimental and epidemiological investigation.^{62–64}

We also observed increased lung cancer risk associated with several pleural and related conditions (eg, pulmonary oedema, empyema and pleural plaques). Prior evidence is sparse or mixed for some of these phenotypes;^{65 66} therefore, these results should be interpreted cautiously and prioritised for external validation, ideally with careful control for imaging intensity and underlying cardiopulmonary comorbidity.

In subgroup analyses, men with respiratory diseases had a higher relative risk of lung cancer than women. This difference may reflect sex-patterned distributions of smoking and other exposures across respiratory conditions rather than a sex-specific biological effect.^{67–70} We also observed higher lung cancer risk among participants with respiratory diseases who reported more frequent baseline alcohol intake. The literature on alcohol and lung cancer is heterogeneous, and in our study, alcohol

intake was measured only at baseline;^{71–73} thus, this association may capture correlated lifestyle patterns rather than a direct causal effect. Given the borderline p value (0.046), the alcohol-related finding should be interpreted as exploratory.

Mediation analyses suggested that leucocyte- and neutrophil-related measures may partially account for some associations between respiratory diseases and lung cancer, although the PM was modest. These markers should be viewed as exploratory signals rather than validated mediators or clinical biomarkers.⁷⁴ In contrast, MR analyses did not support a causal relationship for most respiratory phenotypes. Considering that we applied multiple strategies to reduce reverse causation and confounding in the observational analyses (including excluding lung cancers diagnosed within 1 year after respiratory disease diagnosis, restricting the analytic cohort, multivariable adjustment and sensitivity analyses), nonetheless, residual confounding remains plausible. Consequently, the discrepancy between observational and MR findings may indicate that respiratory diseases function as markers of underlying exposures (eg, smoking, alcohol use, adiposity or other shared determinants) that contribute to inflammation and lung cancer risk, rather than being direct causes themselves.

Because this study is observational and key lifestyle behaviours (including smoking and alcohol use) were assessed only at baseline, the findings regarding lifestyle factors are descriptive and hypothesis-generating rather than prescriptive. Although we observed differences in lung cancer incidence across behaviour categories among participants with respiratory diseases, these associations do not establish causality and do not provide a basis for specific lifestyle or screening recommendations beyond existing public health guidance. Future studies should focus on whether these respiratory associations can improve the precision of current risk-based screening models.

A key strength of this study is the use of a large-scale prospective cohort, enabling evaluation of incidence risks across a broad range of respiratory diseases and lung cancer subtypes. We accounted for multiple covariates, reduced potential reverse causation by excluding lung cancers diagnosed soon after respiratory disease onset and used complementary approaches (mediation and MR) to explore potential biomarkers and causal inference.

Nonetheless, this study has limitations. First, rare respiratory diseases were excluded due to limited case numbers. Second, respiratory disease ascertainment relied strictly on hospital inpatient records, meaning the findings primarily reflect severe episodes. Consequently, milder infections managed in primary care or outpatient settings were not captured, potentially underestimating the total association burden. Third, our grouping strategy focused on the earliest qualifying respiratory diagnosis and did not fully model coexisting respiratory conditions over time. Fourth, covariates were

measured at a single time point and may have changed during follow-up. Fifth, despite Bonferroni correction, chance findings cannot be fully excluded; for some acute upper respiratory infections (URIs), sparse subsequent lung cancer events may yield statistical instability. Importantly, detection and work-up bias remain plausible in this design. Acute infections and pleural effusions often trigger diagnostic imaging, which may lead to the incidental detection of subclinical lung cancers. Moreover, the nature of UK-based coding data makes it difficult to definitively verify the accuracy of infection codes or distinguish whether observed associations reflect true causality, coding errors or coinfections. This potential for detection bias should be considered when interpreting the strong associations observed for these acute conditions. Furthermore, our MR analyses should be regarded as exploratory: stringent instrument selection and F-statistic thresholds meant that only a small number of SNPs (with modest variance explained) were available for several phenotypes, which may produce imprecise or unstable estimates; larger GWAS with expanded sample sizes will be required for more definitive causal inference. Finally, the over-representation of Whites in the UK Biobank cohort may limit generalisability to more ethnically diverse populations,^{75 76} and registry-based linkage data may contain some diagnostic misclassification despite systematic quality assurance.

In conclusion, in this secondary analysis of the UK Biobank, respiratory diseases—including certain acute lower respiratory infections—were associated with an increased subsequent incidence of lung cancer. These findings reflect observed associations and should be interpreted with caution, considering the inherent risk of residual confounding and detection bias in observational cohorts. Mediation analyses suggested that inflammation-related blood markers (notably neutrophils) explained only a modest proportion of these associations. Further studies in independent cohorts and experimental settings are needed to confirm these observations and clarify the underlying mechanisms.^{77 78}

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